BACKTEST PARAMETER COMPARISON

Screening yield, confirmation coverage, and parameter sensitivity

Evaluation window: January 1, 2022 to June 9, 2026 | **Parameter sets:** 192 | **Compact outcomes:** 4,550

Key conclusion: volume-ratio threshold is the dominant driver of screening yield. The broadest configuration selected 92 A-F and 23 A-H securities, while stricter volume settings reduced the candidate pool rapidly. This report compares screening coverage, not realized investment-return quality.

Executive Summary

- The run completed successfully across all 192 parameter combinations with no duplicate outcomes, missing core prices, invalid dates, or high/low inconsistencies.
- A-F produced 3,698 parameter-security outcomes across all 192 parameter sets and 92 unique securities.
- A-H produced 852 outcomes across 132 parameter sets and 23 unique securities; weekly confirmation substantially narrows the candidate universe.
- Lower volume-ratio thresholds and shorter fundamental-history requirements produce the largest candidate pools.
- Return, median return, win rate, and drawdown ranking are not present in the execution log; those metrics should be queried from `backtest_selection_outcome` before selecting a final strategy.

Validated Run Overview

Screen	Outcomes	Sets with results	Unique stocks	Earliest	Latest
A-F	3,698	192	92	2022-01-26	2026-05-27
A-H	852	132	23	2022-02-18	2026-04-10

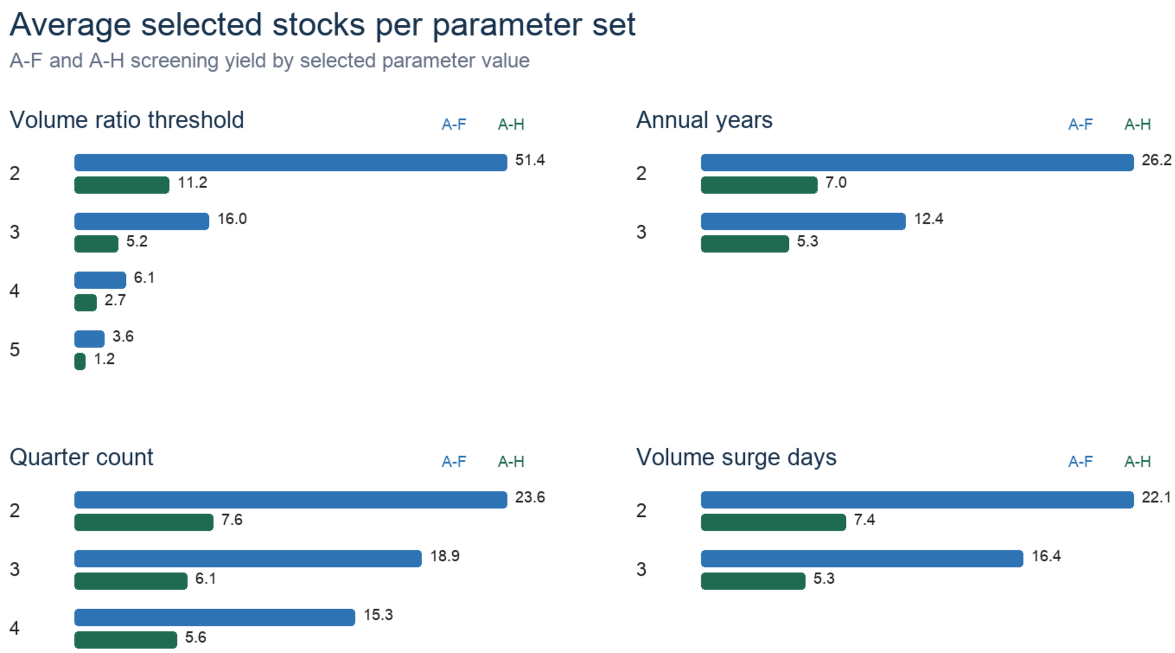

Figure 1. Average screening yield by parameter choice.

Parameter Sensitivity

Factor	Compared values	Avg A-F yield	Avg A-H yield
Annual growth threshold	2 -> 3	19.9 -> 18.6	6.6 -> 6.3
Quarterly growth threshold	2 -> 3	20.1 -> 18.5	6.6 -> 6.3
Annual periods	2 -> 3	26.2 -> 12.4	7.0 -> 5.3
Quarterly periods	2 -> 4	23.6 -> 15.3	7.6 -> 5.6
Volume ratio threshold	2 -> 5	51.4 -> 3.6	11.2 -> 1.2
Volume surge days	2 -> 3	22.1 -> 16.4	7.4 -> 5.3

Interpretation

- Volume ratio is the strongest selectivity lever: average A-F yield falls from 51.4 stocks at 2x volume to 3.6 at 5x; average A-H yield falls from 11.2 to 1.2.
- Requiring three annual periods instead of two roughly halves average A-F yield, from 26.2 to 12.4.
- Increasing the quarterly requirement from two to four periods reduces average A-F yield from 23.6 to 15.3.
- Moving from two to three surge days reduces average A-F yield from 22.1 to 16.4, a meaningful but less severe reduction than volume ratio.

Highest-Coverage Parameter Sets

ID	Growth A/Q	Periods A/Q	Volume	A-F	A-H	Total
1	2/2	2/2	2x/2d	92	23	115
97	3/2	2/2	2x/2d	88	21	109
49	2/3	2/2	2x/2d	85	22	107
145	3/3	2/2	2x/2d	81	20	101
9	2/2	2/3	2x/2d	79	20	99
105	3/2	2/3	2x/2d	76	18	94
2	2/2	2/2	2x/3d	78	16	94
57	2/3	2/3	2x/2d	73	17	90
50	2/3	2/2	2x/3d	73	16	89
98	3/2	2/2	2x/3d	74	14	88

Recommended Comparison Set

Role	ID	Configuration	A-F / A-H	Use
Broad discovery	1	2%/2%, 2 annual, 2 quarters, 2x volume, 2 surge days	92 / 23	Largest sample; suitable baseline.
Moderate volume	3	2%/2%, 2 annual, 2 quarters, 3x volume, 2 surge days	36 / 13	Meaningfully stricter while retaining A-H coverage.
Strict volume	7	2%/2%, 2 annual, 2 quarters, 5x volume, 2 surge days	9 / 2	Small sample; useful only as a high-conviction comparator.
Durability focus	41	2%/2%, 3 annual, 4 quarters, 2x volume, 2 surge days	33 / 7	Longer fundamental history with usable sample size.

Decision Guidance

- Do not label the highest-coverage parameter set as the best-performing investment strategy. Coverage and return quality answer different questions.
- Require a minimum sample size before ranking return performance. A practical initial threshold is at least 20 A-F outcomes or at least 10 A-H outcomes.

- Compare mean and median return together. A large gap often indicates that a few extreme winners dominate the average.
- Use win rate and maximum drawdown alongside returns; parameter sets with high average returns but severe drawdowns may be unsuitable.
- Evaluate A-F and A-H separately. A-H is a stricter weekly-confirmed strategy and should not be pooled with A-F.

Required Return-Performance Query

To complete an investment-performance ranking, export the following aggregate query from Supabase and use it for a second report:

```
SELECT p.parameter_set_id, p.parameter_set_name, o.screen_type, COUNT(*) AS sample_size,
ROUND(AVG(o.return_pct), 2) AS avg_return_pct,
ROUND(PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY o.return_pct)::numeric, 2) AS median_return_pct,
ROUND(100.0 * COUNT(*) FILTER (WHERE o.return_pct > 0) / COUNT(*), 2) AS win_rate_pct,
ROUND(AVG(o.max_drawdown_pct), 2) AS avg_drawdown_pct,
ROUND(AVG(o.max_return_pct), 2) AS avg_max_return_pct
FROM backtest_parameter_set p JOIN backtest_selection_outcome o USING (parameter_set_id)
GROUP BY p.parameter_set_id, p.parameter_set_name, o.screen_type
ORDER BY o.screen_type, sample_size DESC;
```

Scope and Limitations

- This report uses the validated execution log and compares selection yield, not realized portfolio returns.
- The same security may appear in many parameter sets, so outcome rows are not independent observations.
- Fundamental histories use `datadate <= selected_date`; actual public filing-availability dates are unavailable, so some look-ahead-bias risk remains.
- No transaction costs, liquidity constraints, holding rules, or overlapping-position controls are included.